

Muhammad Abdullah Goher

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Education

University of Pennsylvania

Aug 2023 – May 2027

B.S.E. & M.S.E. in Computer Science (Submatriculation), Concentration in Artificial Intelligence

Philadelphia, PA

GPA: 3.72 (B.S.E.) / 3.88 (M.S.E.) | **Coursework:** Big Data Analytics, Applied Machine Learning, Probability, Statistical Inference, Linear Algebra, Databases, Data & Analysis for Decisions, Algorithms

Experience

Computational Social Science Lab (Prof. Duncan Watts) – Data Science Research Assistant

May 2025 – Dec 2025

Large-scale human-mobility analytics for disaster & tourism planning

Philadelphia, PA

- Analyzed **multi-terabyte** GPS mobility datasets (Meta Crisis Population, Factori, SafeGraph) using PySpark on AWS EMR to compute high-resolution crowd-density and movement trends for 2026 FIFA World Cup planning.
- Visualized insights through heat maps and geospatial plots of GPS pings and POI polygons with Matplotlib and GeoPandas.
- Built a Generative-AI (RAG) extraction pipeline pulling quantitative mobility metrics from **100+** scholarly papers.

AeternalLabs, PBC – Founding Engineer

May 2026 – Present

Runtime AI-governance & fairness-analytics platform (first engineering hire)

Remote

- Built data and evaluation infrastructure measuring demographic bias across **18+ models** from 8 providers, implementing a 22-dimension scoring vector with statistical comparison tests.
- Designed a PostgreSQL analytics store (SQLAlchemy) with integrity hashing capturing every model run for reproducible, auditable analysis.

Clab AI – Data Analyst Intern

May 2024 – Aug 2024

AI-powered college-application assistant used by 100+ students

Hybrid – Nashville, TN

- Performed extensive EDA, data cleaning, and visualization on a dataset of **100+** universities, surfacing trends in financial-aid awards and admissions patterns.
- Built hyperparameter-tuned Random Forest / Linear Regression models predicting aid eligibility at **82% R^2** on validation.

PennAdapt – Machine Learning Engineer

Sep 2024 – Present

Penn Assistive Devices & Prosthetic Technologies

Philadelphia, PA

- Built a computer-vision model (CNN + Vision Transformer, PyTorch) classifying **26 classes** of surgical tools at **90%+** accuracy on a 1,500+ image dataset curated with Penn Medicine.

Projects

Steam Dataset EDA & Modeling | *Python, pandas, scikit-learn, XGBoost, LightGBM*

- Cleaned, preprocessed, and visualized a **110k+** game dataset; trained Random Forest, XGBoost, and LightGBM models to predict commercial success, with EDA in Matplotlib, Seaborn, and Plotly.

News Source Classification | *PyTorch, HuggingFace Transformers, scikit-learn*

- Co-built (team of 3) a headline classifier; scaled the dataset **3,800** → **17,900** headlines via custom scraping, fine-tuning RoBERTa to **91.75% accuracy** (+25 pts over a TF-IDF baseline).

West Coast EV Explorer | *PostgreSQL, PostGIS, Node.js, Express, React*

- Designed a normalized schema and materialized views over a **100k+** row database; built APIs for zip-code EV lookups and PostGIS nearest-charger search.

Technical Skills

Data Science: pandas, NumPy, PySpark, SQL, scikit-learn, hypothesis testing, feature engineering, EDA, A/B analysis

Visualization: Matplotlib, Seaborn, Plotly, GeoPandas, Tableau-style dashboards, Jupyter

ML / AI: PyTorch, XGBoost/LightGBM, HuggingFace Transformers, NLP, fine-tuning (RLHF), RAG, Vision Transformers

Languages & Tools: Python, R, SQL, C, Java, JavaScript; PostgreSQL, MongoDB, Docker, AWS (S3, EMR), Git, Linux